



DARE-Arbo Surveillance Study Design Report



Protocol-oriented summary generated by the DARE-Arbo Designer. Planning coverage is not a completed risk-of-bias assessment.

Planning framework complete

Coverage floor: 100% | Analysis outputs: 8

1. Surveillance objective and study architecture

Planned study title	District integrated arbovirus surveillance study
Target population	General febrile population attending sentinel facilities
Virus coverage	Multiplex / multiple viruses
Arbovirus target(s)	DENV; CHIKV
Surveillance mode	Hybrid active + passive surveillance
Surveillance architecture	Sentinel facility surveillance
Planned recruited sample size	1200
Sampling and recruitment	Multistage probability sampling
Endpoint design	Mixed endpoints
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Active surveillance source	Scheduled community fever screening
Passive surveillance source	Routine febrile presentations at sentinel facilities
Stream analysis	Separate active and passive estimates
Integration prespecified	Yes

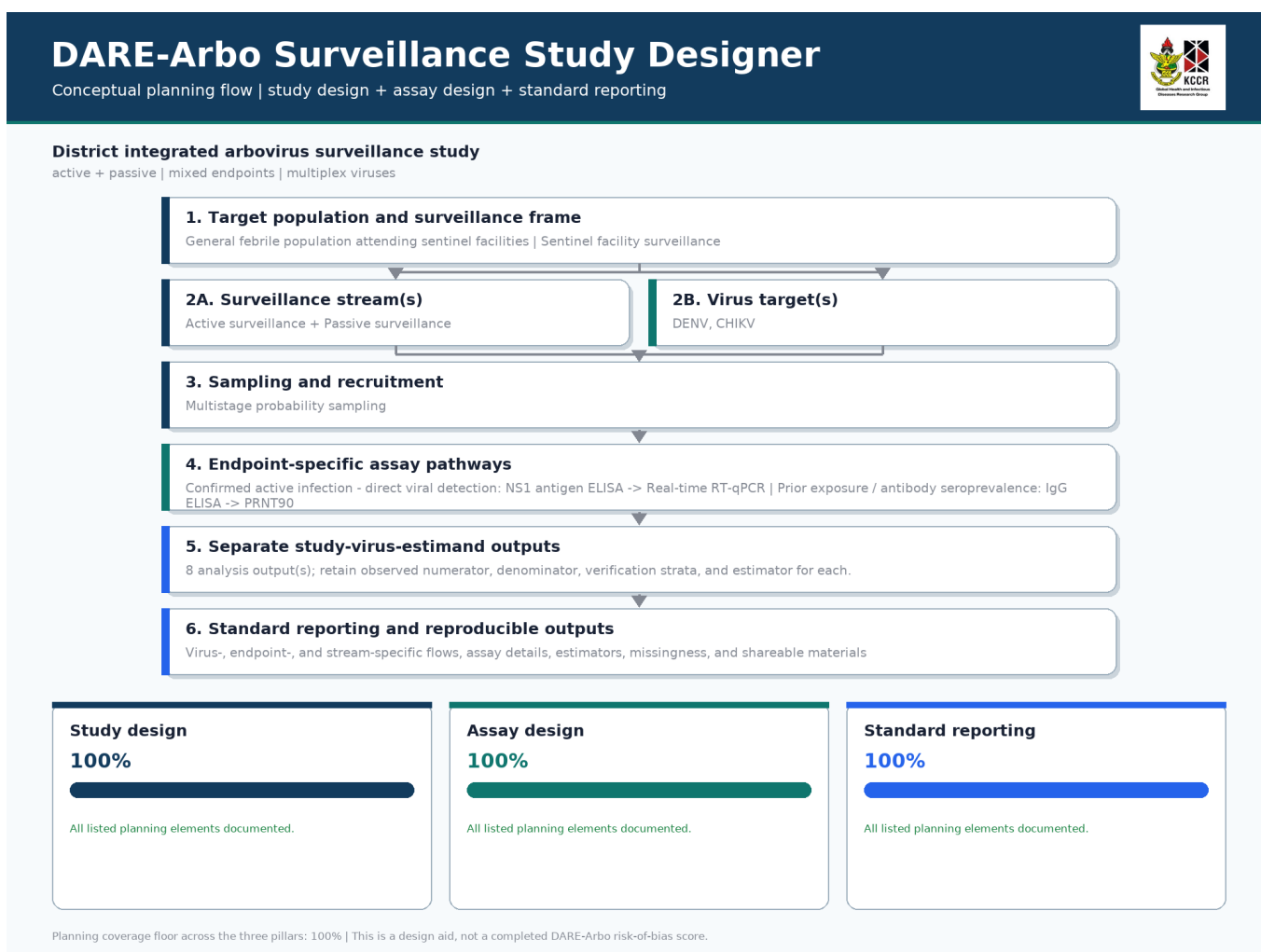
2. Planning balance

Planning pillar	Coverage	Outstanding elements
Study design	100%	Complete
Assay design	100%	Complete
Standard reporting	100%	Complete

Planning priorities

No outstanding planning elements were identified by the Designer checklist.

3. Surveillance design flow



The figure links the declared population, surveillance stream, endpoint-specific assay pathway, and reporting output.

4. Study-virus-estimand analysis plan

Analysis unit 1: Active surveillance | DENV | Confirmed active infection - direct viral detection

Testing strategy	Molecular testing of all screening positives
Primary / screening assay	NS1 antigen ELISA
Confirmatory assay(s)	Real-time RT-qPCR
Observed numerator	Molecular positive among all screening positives
Estimator denominator	Total tested (complete screening-algorithm entry population)
Summary estimator	Tier B (confirmed active infection)

Analysis unit 2: Active surveillance | DENV | Prior exposure / antibody seroprevalence

Testing strategy	ELISA/RDT only
Primary / screening assay	IgG ELISA
Confirmatory assay(s)	PRNT90
Observed numerator	Primary positive
Estimator denominator	Total tested
Summary estimator	Apparent prevalence

Analysis unit 3: Active surveillance | CHIKV | Confirmed active infection - direct viral detection

Testing strategy	Molecular testing of all screening positives
Primary / screening assay	NS1 antigen ELISA
Confirmatory assay(s)	Real-time RT-qPCR
Observed numerator	Molecular positive among all screening positives
Estimator denominator	Total tested (complete screening-algorithm entry population)
Summary estimator	Tier B (confirmed active infection)

Analysis unit 4: Active surveillance | CHIKV | Prior exposure / antibody seroprevalence

Testing strategy	ELISA/RDT only
Primary / screening assay	IgG ELISA
Confirmatory assay(s)	PRNT90
Observed numerator	Primary positive
Estimator denominator	Total tested
Summary estimator	Apparent prevalence

Analysis unit 5: Passive surveillance | DENV | Confirmed active infection - direct viral detection

Testing strategy	Molecular testing of all screening positives
Primary / screening assay	NS1 antigen ELISA
Confirmatory assay(s)	Real-time RT-qPCR
Observed numerator	Molecular positive among all screening positives
Estimator denominator	Total tested (complete screening-algorithm entry population)
Summary estimator	Tier B (confirmed active infection)

Analysis unit 6: Passive surveillance | DENV | Prior exposure / antibody seroprevalence

Testing strategy	ELISA/RDT only
Primary / screening assay	IgG ELISA
Confirmatory assay(s)	PRNT90
Observed numerator	Primary positive
Estimator denominator	Total tested
Summary estimator	Apparent prevalence

Analysis unit 7: Passive surveillance | CHIKV | Confirmed active infection - direct viral detection

Testing strategy	Molecular testing of all screening positives
Primary / screening assay	NS1 antigen ELISA
Confirmatory assay(s)	Real-time RT-qPCR
Observed numerator	Molecular positive among all screening positives
Estimator denominator	Total tested (complete screening-algorithm entry population)
Summary estimator	Tier B (confirmed active infection)

Analysis unit 8: Passive surveillance | CHIKV | Prior exposure / antibody seroprevalence

Testing strategy	ELISA/RDT only
Primary / screening assay	IgG ELISA
Confirmatory assay(s)	PRNT90
Observed numerator	Primary positive
Estimator denominator	Total tested
Summary estimator	Apparent prevalence

5. Standard reporting commitments

Reporting commitment	Status
Observed endpoint numerator and denominator	Planned
Participant/specimen and verification-stratum flow	Planned
Assay manufacturer, protocol, cutoffs, controls, and interpretation	Planned
Nonresponse, exclusions, missing specimens, and incomplete testing	Planned
Estimator, weights, standardization, and assay adjustment	Planned
Protocol, codebook, analysis code, or machine-readable outputs	Planned
Endpoint-specific results retained separately	Planned
Virus-specific and co-detection reporting	Planned

6. Protocol handoff note

This report documents a planned design. It does not replace a protocol, statistical analysis plan, ethics review, laboratory SOP, or completed DARE-Arbo risk-of-bias assessment. Before implementation, resolve every planning priority and verify that each virus-endpoint-stream analysis unit has a matched observed numerator, denominator, assay pathway, and estimator.